

Deoliana Matilde Tilman Magalhaes

FRESH GRADUATE. BIOMEDICAL ENGINEERING

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PROFILE SUMMARY

Biomedical Engineering graduate (Bachelor degree at Korea University, 2026) with hands-on research experience across the hardware and software sides of medical imaging from transducer simulation and RF signal processing to deep learning-based image reconstruction and classification. Skilled in Python (PyTorch, scikit-learn), MATLAB, and Verasonics systems, with experience validating and analyzing complex clinical datasets including MRI/ultrasound imaging and spectroscopic data. Seeking to apply this research background to AI-driven medical imaging and diagnostic development.

PROFESSIONAL EXPERIENCE

Research Intern — Medical Imaging System Laboratory <i>Department of Biomedical Engineering, Korea University, Seoul</i>	Jul 2025 – Mar 2026
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- Conducted research on ultrasound imaging and deep learning-based beamforming for image reconstruction
- Applied Python and MATLAB for signal processing, data analysis, and imaging simulations using Verasonics systems
- Processed and analyzed large-scale medical imaging datasets for model development and evaluation
- Developed simulation experiments and validated results for imaging performance improvement
- Presented research progress and findings in weekly lab meetings

Research Intern — Nano Bioplasmonic Laboratory <i>Korea University × KIST, Seoul</i>	Sep 2024 – Mar 2025
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- Conducted Raman spectroscopy experiments and analyzed spectral data using machine learning models (Random Forest, SVM, CNN) for classification tasks
- Supported experimental setup, data collection, and laboratory documentation

Head of Public Affairs — Student Association <i>Timor-Leste Embassy Community, Seoul</i>	Aug 2024 – Mar 2026
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- Coordinated events and programs connecting Timor-Leste students with embassy and external partners
- Assisted in organizing cultural and community activities promoting Timor-Leste in Korea

EDUCATION

Bachelor of Biomedical Engineering
Korea University, Seoul, South Korea
Mar 2022 – Feb 2026

Natural Science Program (Secondary School)
Sta. Madalena de Canossa, Dili, Timor-Leste
Jan 2017 – Dec 2019

LANGUAGES

Tetun	Native
English	Professional
Korean	TOPIK 4
Indonesian	Fluent

TECHNICAL SKILLS

Programming & ML — Python (PyTorch, scikit-learn, RamanSPy, pandas), MATLAB (signal processing, simulation)

Hardware & Simulation — Verasonics, transducer array simulation, RF data acquisition

Data & Dev Tools — VS Code, Jupyter Lab, Git/GitHub, Microsoft Excel

3D Modeling — Rhino, Fusion 360

Visualization & Presentation — PowerPoint, Canva

LABORATORY & RESEARCH SKILLS

- Medical imaging data processing and simulation (ultrasound)
- Raman spectroscopy and basic cell culture
- Experimental data collection, analysis, and validation
- Research documentation and reporting
- Strong analytical and problem-solving skills
- Adaptable and quick to learn new systems and technologies

ADDITIONAL EXPERIENCE & ACHIEVEMENTS

Global Korea Scholarship (GKS) Recipient — NIIED Awarded prestigious government scholarship for undergraduate studies in Biomedical Engineering.	2021 – 2026
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Global Peace Foundation — Volunteer & Program Organizer - Supported the organization of international summer camp programs for diverse participants - Served as Master of Ceremony (MC) for major event sessions	Aug 2024
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Korean Dream Crew — Member, Global Youth Program - Participated in leadership and cross-cultural collaboration programs - Recognized three times for outstanding team content creation	Feb 2024 – Aug 2024
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ASEAN Science, Technology & Innovation Cooperation (STIC) - Completed online training in Big Data and Cybersecurity - Strengthened understanding of data systems and digital technologies	Sep 2023 – Jan 2024
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Presentation Contest Winner — Korea University Awarded for outstanding academic presentation.	Nov 2022
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